

When Should Actuaries Trust Neural Networks?

A Comprehensive Benchmark of Classical and Neural Mortality Models with a Hybrid LC-ResNet and Practical Decision Framework

Amine Manai

Master 1 Actuariat, Le Mans Université

`amine.manai@esprit.tn`

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Abstract

We present a comprehensive benchmark comparing 15 mortality forecasting models—8 classical (implemented from scratch), 6 neural network architectures, and an original hybrid—across 8 countries from the Human Mortality Database (1950–2023). Using rolling-origin validation with actuarial metrics (life expectancy e_0 , annuity values $\ddot{a}_{65}$), we address practical questions that existing studies leave unanswered: which model to use with short historical data, which is most robust to mortality shocks (COVID-19), and when the added complexity of neural networks is justified.

Our main contributions are: (1) a practical decision framework mapping data characteristics to optimal model choice; (2) **LC-ResNet**, a hybrid model combining a Poisson Lee-Carter skeleton with a neural residual correction subject to horizon-dependent shrinkage $\exp(-\lambda h)$, ensuring stable long-horizon extrapolation while capturing short-term non-linearities; and (3) an actuarial case study quantifying model risk in EUR on a French annuity portfolio under Solvency II.

Key findings: the random walk with drift remains a strong baseline at all horizons; neural κ_t replacements (LSTM, GRU) outperform classical Lee-Carter at short horizons ($h \leq 5$) but offer diminishing gains beyond; LC-ResNet ranks among the top models for young ages while preserving interpretability; and the inter-model spread on a 1,000-life annuity portfolio reaches 6.3 million EUR, underscoring the materiality of model risk.

Keywords: mortality forecasting, Lee-Carter, neural networks, LSTM, model risk, longevity risk, Solvency II, annuity pricing, decision framework.

JEL: C45, C52, C53, G22, J11.

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1 Introduction

Longevity risk—the risk that policyholders live longer than expected—is a primary concern for life insurers and pension funds. Under Solvency II, the capital requirement for longevity risk ($\text{SCR}_{\text{longevity}}$) typically represents 5–15% of the best estimate of annuity provisions, making the choice of mortality model a capital allocation decision rather than a purely academic exercise.

The Lee-Carter family of models (Lee and Carter, 1992) has dominated actuarial practice for over three decades. In parallel, neural network approaches have shown promising results in recent literature (Richman and Wüthrich, 2021; Nigri et al., 2019; Perla et al., 2021). Yet no study provides a practical guide for actuaries to choose between classical and neural models based on their specific context: data availability, forecast horizon, age segment of interest, and regulatory constraints.

This paper fills that gap by benchmarking 15 models across 8 countries with rolling-origin validation, then distilling the results into a decision framework and quantifying the financial consequences through an actuarial case study.

1.1 Contributions

1. A **comprehensive benchmark** of 15 models (8 classical, 6 neural, 1 hybrid) on 8 countries from the Human Mortality Database, evaluated with both statistical and actuarial metrics under rolling-origin validation.
2. **LC-ResNet**, an original hybrid model combining a Poisson Lee-Carter skeleton with a neural residual network and horizon-dependent shrinkage, designed to preserve interpretability while capturing short-term non-linearities.
3. A **practical decision framework** mapping data characteristics (history length, forecast horizon, age segment, interpretability needs) to optimal model choice, based on scenario-specific evaluations.
4. An **actuarial case study** pricing a French annuity portfolio and quantifying model risk in EUR, including a Solvency II longevity shock analysis.

1.2 Related work

Lee and Carter (1992) introduced the seminal Lee-Carter model; subsequent extensions include Lee-Miller (Lee and Miller, 2001), Booth-Maindonald-Smith (Booth et al., 2002), the Poisson formulation of Brouhns et al. (2002), the two-factor CBD model for older ages (Cairns et al., 2006), and the functional data approach of Hyndman and Ullah (2007). On the neural side, Nigri et al. (2019) applied LSTMs to the κ_t index, while Richman and Wüthrich (2021) proposed a feed-forward network with entity embeddings for multi-population mortality. Perla et al. (2021) provided a systematic review of deep learning methods for mortality forecasting.

Our work differs by (i) benchmarking all these approaches under identical conditions with rolling-origin validation, (ii) evaluating on actuarial rather than purely statistical metrics, and (iii) proposing a hybrid model and a practical decision framework.

2 Data

We use data from the **Human Mortality Database** (HMD; www.mortality.org) for 8 countries: France (FRATNP), England & Wales (GBRTENW), United States (USA), Japan (JPN), Italy (ITA), Spain (ESP), Sweden (SWE), and Netherlands (NLD). For each country, we extract 1×1 (single-year age, single calendar year) central death rates $m_{x,t}$ and exposures $E_{x,t}$ for ages 0–100 and years 1950–2023, total population (both sexes combined).

The period 1950–2023 is chosen to include sufficient historical depth for long-horizon evaluation and to encompass the COVID-19 mortality shock (2020–2023), which serves as a natural out-of-sample stress test.

Zero or missing rates are floored at 10^{-6} before taking logarithms.

3 Models

All classical models are implemented from scratch in Python using NumPy and SciPy; neural models use PyTorch. All share a common `MortalityModel` interface (`fit`, `forecast`, `simulate`) to ensure evaluation consistency.

3.1 Classical models

Lee-Carter (LC). [Lee and Carter \(1992\)](#): $\ln m_{x,t} = a_x + b_x \kappa_t + \varepsilon_{x,t}$, estimated by SVD with identifiability constraints $\sum_x b_x = 1$, $\sum_t \kappa_t = 0$, and κ_t re-estimated on observed deaths via constrained optimization. Forecast: κ_t follows a random walk with drift (RWD).

Lee-Miller (LM). [Lee and Miller \(2001\)](#): extends LC by adjusting κ_t so that the model-implied e_0 matches the observed e_0 each year, solved via Brent’s root-finding algorithm.

Booth-Maindonald-Smith (BMS). [Booth et al. \(2002\)](#): selects the optimal fitting period by testing the linearity of κ_t over sub-periods, then applies standard LC on the selected window.

Poisson Lee-Carter (PLC). [Brouhns et al. \(2002\)](#): $D_{x,t} \sim \text{Poisson}(E_{x,t} \cdot \exp(a_x + b_x \kappa_t))$, estimated by maximum likelihood via alternating unidimensional Newton-Raphson iterations.

Cairns-Blake-Dowd (CBD). [Cairns et al. \(2006\)](#): a two-factor model for ages 60–100: $\text{logit } q_{x,t} = \kappa_t^{(1)} + \kappa_t^{(2)}(x - \bar{x})$, with bivariate RWD on $(\kappa_t^{(1)}, \kappa_t^{(2)})$.

Hyndman-Ullah (HU). [Hyndman and Ullah \(2007\)](#): functional data analysis with penalized B-spline smoothing, K -component functional PCA, and ARIMA forecasting on each score series. We use $K = 6$ components.

Baselines. Random walk with drift (RWD) on age-specific $\ln m_{x,t}$, and frozen rates (last observed year projected forward).

3.2 Neural models

All neural models share identical training hyperparameters for benchmark fairness: 200 epochs, patience 20 (early stopping on validation loss), learning rate 10^{-3} (Adam, weight decay 10^{-4}), seed 42.

Neural κ_t models (LSTM, GRU, Bi-LSTM, Transformer). The Lee-Carter skeleton (a_x, b_x) is preserved; only the κ_t forecast is replaced by a recurrent or attention-based architecture (2 layers, hidden size 64, lookback 20 years). This retains the interpretable age structure while potentially capturing non-linear temporal patterns.

FFNN with embeddings. Inspired by [Richman and Wüthrich \(2021\)](#): an age embedding (dimension 8) concatenated with a continuous normalized year feature, fed through a 3-layer MLP (128–64–32). Outputs $\ln m_{x,t}$ directly.

CNN on the mortality surface. Treats the (age $\times$ year) matrix as a 2D image; three convolutional layers (16, 32, 64 filters, kernel size 3) with adaptive average pooling predict the next year’s mortality profile from a sliding window.

3.3 LC-ResNet: a hybrid model

Our original contribution combines a Poisson Lee-Carter skeleton with a neural residual correction:

1. **Fit** a Poisson Lee-Carter to obtain $\hat{a}_x, \hat{b}_x, \hat{\kappa}_t$.
2. **Compute structured residuals:** $r_{x,t} = \ln m_{x,t} - (\hat{a}_x + \hat{b}_x \hat{\kappa}_t)$.
3. **Train a residual network** f_θ on features $[\tilde{x}, \tilde{t}, \tilde{b}_x, \tilde{\kappa}_t]$ (normalized) with architecture MLP(64, 32) to predict $r_{x,t}$.
4. **Forecast with shrinkage:** for horizon h ,

$$\widehat{\ln m_{x,t+h}} = \underbrace{\hat{a}_x + \hat{b}_x \hat{\kappa}_{t+h}}_{\text{LC skeleton}} + \underbrace{e^{-\lambda h} \cdot f_\theta(\tilde{x}, \tilde{t} + h, \tilde{b}_x, \tilde{\kappa}_{t+h})}_{\text{shrunk neural correction}} \quad (1)$$

The shrinkage factor $e^{-\lambda h}$ (with $\lambda = 0.1$) is the key design choice: at $h = 1$, $e^{-0.1} \approx 0.90$, so the neural correction is nearly fully active; at $h = 20$, $e^{-2} \approx 0.14$, so the model reverts to stable LC extrapolation. This prevents the common failure mode of neural mortality models—extrapolation instability at long horizons—while preserving short-term flexibility.

The parameters a_x, b_x, κ_t remain fully interpretable and accessible for actuarial reporting, unlike purely neural approaches.

4 Evaluation Protocol

4.1 Rolling-origin validation

We use rolling-origin cross-validation adapted for time series: the model is trained on years $[1950, T_0]$, then evaluated on the subsequent h years. The origin T_0 ranges from

1990 to 2018 in steps of 2, and horizons $h \in \{1, 5, 10, 20\}$ years. The model is fully re-trained at each origin—no incremental learning.

4.2 Metrics

We report both statistical and actuarial metrics:

- **RMSE on $\ln m_{x,t}$:** root mean squared error on log central death rates.
- **RMSE on e_0 :** error on period life expectancy at birth (in years).
- **RMSE on $\ddot{a}_{65}$:** error on the whole-life annuity-due factor at age 65 (interest rate $i = 2\%$).

Actuarial metrics are computed only when the model covers the full age grid (0–100); the CBD model, restricted to ages 60–100, is evaluated on $\ln m_{x,t}$ only.

4.3 Statistical testing

We use the Diebold-Mariano test (Diebold and Mariano, 1995) with the Harvey-Leybourne-Newbold correction (Harvey et al., 1997) for multi-step autocorrelation and small-sample adjustment, referencing a Student’s t distribution.

4.4 Scenario-specific evaluations

- **Short history:** training truncated to 20, 30, or 50 years to assess data-efficiency.
- **Mortality shock:** training ends in 2019, evaluation on 2020–2023 (COVID-19).
- **Age groups:** separate evaluation on young (0–19), working (20–64), and elderly (65–100) ages.

5 Results

The full benchmark produces 29,536 observations across 15 models, 8 countries, 15 rolling origins, 7 horizons, and multiple metrics. We report cross-country averages; country-level results are available in the online repository.

5.1 Main results: RMSE on $\ln m_{x,t}$

Table 1 reports the average RMSE on $\ln m_{x,t}$ by model and horizon.

Key observations.

1. The **random walk with drift is a remarkably strong baseline**, outperforming most classical LC variants and several neural models at all horizons.
2. **Neural κ_t models (LSTM, GRU)** improve over classical Lee-Carter at short horizons ($h \leq 5$) but offer diminishing or negative gains at $h = 20$.

Table 1: RMSE on $\ln m_{x,t}$, averaged across 8 countries (lower is better). Best in **bold**.

Model	$h = 1$	$h = 5$	$h = 10$	$h = 20$
<i>Classical</i>				
Lee-Carter	0.157	0.207	0.199	0.248
Lee-Miller	0.146	0.196	0.187	0.232
BMS	0.157	0.207	0.199	0.248
Poisson LC	0.193	0.244	0.231	0.272
CBD (60+ only)	0.081	0.126	0.095	0.120
Hyndman-Ullah	0.318	0.351	0.329	0.355
<i>Baselines</i>				
Random walk	0.103	0.155	0.146	0.203
Frozen rates	0.103	0.164	0.186	0.314
<i>Neural</i>				
LSTM κ_t	0.132	0.182	0.180	0.263
GRU κ_t	0.132	0.183	0.182	0.274
Bi-LSTM κ_t	0.133	0.188	0.195	0.288
Transformer κ_t	0.141	0.207	0.225	0.327
FFNN embeddings	0.233	0.286	0.290	0.368
CNN surface	0.151	0.221	0.240	0.448
<i>Hybrid</i>				
LC-ResNet	0.122	0.182	0.184	0.247

Note: CBD is evaluated only on ages 60–100, making its RMSE not directly comparable to full-grid models.

3. **LC-ResNet** achieves the best non-CBD RMSE at $h = 1$ (0.122) and remains competitive at all horizons. At $h = 20$, it matches Lee-Miller (0.247 vs. 0.232), demonstrating the effectiveness of the shrinkage mechanism.
4. The **Transformer** and **CNN** are the weakest neural architectures, suffering from extrapolation instability at long horizons (RMSE 0.327 and 0.448 at $h = 20$).
5. **Hyndman-Ullah** and **FFNN** underperform across all horizons in this benchmark.

5.2 Actuarial metrics

Table 2 reports RMSE on e_0 (in years) and on $\ddot{a}_{65}$ (dimensionless).

Table 2: Actuarial metrics, averaged across 8 countries.

Model	RMSE e_0 (years)		RMSE $\ddot{a}_{65}$	
	$h = 5$	$h = 20$	$h = 5$	$h = 20$
Lee-Carter	0.495	0.832	0.212	0.397
Lee-Miller	0.386	0.742	0.228	0.398
Poisson LC	0.444	0.826	0.215	0.403
Hyndman-Ullah	0.486	0.694	0.221	0.437
Random walk	0.383	0.764	0.192	0.395
LSTM κ_t	0.807	2.059	0.432	1.017
GRU κ_t	0.815	2.181	0.429	1.065
LC-ResNet	0.421	0.846	0.207	0.403

A striking finding is that **neural κ_t models degrade sharply on actuarial metrics** at long horizons: the LSTM’s e_0 RMSE at $h = 20$ is 2.06 years vs. 0.74 for Lee-Miller. This is because small systematic biases in κ_t forecasts compound through the life table. LC-ResNet avoids this trap thanks to its shrinkage mechanism: at $h = 20$, it performs comparably to the classical LC family on e_0 and $\ddot{a}_{65}$.

5.3 Scenario results

Short history. Table 3 shows that with only 20 years of training data, model differences narrow considerably. Frozen rates and LC-ResNet perform best; neural κ_t models converge to the LC baseline because their recurrent networks lack sufficient training sequences.

Age groups. LC-ResNet ranks first for young ages (0–19) with RMSE 0.310, suggesting that the neural residual correction captures non-linearities in infant and child mortality that the linear $b_x\kappa_t$ structure misses. For elderly ages (65–100), the random walk and Hyndman-Ullah perform best, while CBD—despite being designed for this segment—underperforms relative to models that benefit from information in younger ages.

6 Decision Framework

Based on the benchmark results, we propose the following model selection guidelines:

Table 3: RMSE $\ln m_{x,t}$ with truncated training history (averaged across countries and horizons).

Model	20 yr	30 yr	50 yr
Frozen rates	0.296	–	0.296
LC-ResNet	0.317	–	0.308
Lee-Carter	0.322	–	0.319
LSTM κ_t	0.321	–	0.317
Random walk	0.328	–	0.307
Hyndman-Ullah	0.485	–	0.480

Table 4: Practical decision framework for model selection.

Context	Recommended	Rationale
Short history (< 25 yr)	RWD	Neural networks lack training data; RWD is robust
Elderly ages (65+), regulatory	CBD	Designed for 60+; standard among regulators
Short horizon (≤ 5 yr)	LC-ResNet	Neural correction most active; best at $h = 1$
Medium horizon (5–10 yr)	Poisson LC	Principled likelihood; stable
Long horizon (> 10 yr)	Lee-Miller	Proven stability; e_0 -adjusted
Interpretability required	LC-ResNet	LC skeleton is accessible; correction is bounded

The key insight is that **no single model dominates across all scenarios**. The random walk is surprisingly competitive, and neural κ_t replacements are only justified when the training history is long (≥ 50 years) and the forecast horizon is short (≤ 5 years).

7 Actuarial Case Study

We translate the benchmark results into financial terms using a stylized French annuity portfolio.

7.1 Setup

- 1,000 pensioners aged 65
- Annual payment: 12,000 EUR per person
- Discount rate: $i = 2\%$ (risk-free)
- Mortality projection: each model’s 10-year-ahead forecast

7.2 Annuity pricing

Table 5: Annuity pricing results by model (French mortality data).

Model	$\ddot{a}_{65}$	Provision/person (EUR)	Total (MEUR)
Lee-Carter	18.338	220,050	220.05
Poisson LC	18.323	219,873	219.87
LC-ResNet	18.268	219,217	219.22
Random walk	18.236	218,826	218.83
CBD	17.816	213,793	213.79
Model risk spread			6.26

The inter-model spread on total provisions reaches **6.26 million EUR**—a material amount driven primarily by the CBD model, which projects higher mortality (lower $\ddot{a}_{65}$) for the elderly segment. Excluding CBD, the spread among full-grid models is 1.22 million EUR.

7.3 Solvency II longevity shock

Table 6 shows the impact of a standard Solvency II longevity shock (-20% on mortality rates, i.e., policyholders live longer):

The $\text{SCR}_{\text{longevity}}$ ranges from 11.52 MEUR (Lee-Carter) to 12.33 MEUR (CBD), a spread of 0.81 MEUR—approximately 7% of the SCR itself. This demonstrates that model choice affects not only best-estimate provisions but also regulatory capital.

The SCR as a percentage of the base provision is remarkably stable across models (5.23–5.77%), suggesting that the shock multiplier is robust to model specification, even if the base levels differ.

Table 6: Solvency II longevity shock analysis.

Model	Base (MEUR)	Shocked (MEUR)	SCR _{long.} (MEUR)
Lee-Carter	220.05	231.57	11.52
Poisson LC	219.87	231.39	11.52
LC-ResNet	219.22	230.80	11.58
Random walk	218.83	230.54	11.71
CBD	213.79	226.12	12.33
SCR range			0.81

8 Discussion

The random walk puzzle. The strongest finding of this benchmark is the competitiveness of the age-specific random walk with drift. It outperforms all classical LC variants and most neural models at horizons 5–20. This aligns with Booth (2006) and challenges the assumption that structural models necessarily outperform naïve baselines.

Neural networks: promise and peril. Neural κ_t replacements (LSTM, GRU) show genuine improvement at short horizons ($h \leq 5$), where the recurrent architecture captures recent trend changes that the linear drift misses. However, they degrade at long horizons, and this degradation is amplified on actuarial metrics because e_0 and $\ddot{a}_{65}$ are non-linear functions of the entire mortality surface. The Transformer and CNN architectures perform worst, likely due to limited training data relative to model complexity.

LC-ResNet: bounded flexibility. The hybrid model achieves a favorable trade-off: it captures short-term residual structure (ranking first for young ages) while reverting to stable LC extrapolation at long horizons via the shrinkage mechanism. The interpretable LC skeleton (a_x, b_x, κ_t) remains accessible for regulatory reporting.

Limitations.

- All models are single-population; multi-population extensions (Li-Lee) are not considered.
- Cohort effects are not explicitly modeled (Renshaw-Haberman), though the LC-ResNet residual network may implicitly capture some.
- The neural models use a single seed (42) rather than ensemble averaging, which may underestimate their potential.
- HMD data ends in 2023; post-COVID stabilization effects remain uncertain.

9 Conclusion

This benchmark provides an empirical answer to the question “when should actuaries trust neural networks for mortality forecasting?”: at short horizons with long historical data, when the added complexity is justified by practical gains. For most actuarial applications—long-horizon projections for provisioning and capital—the classical Lee-Carter family and even the simple random walk remain competitive or superior.

The LC-ResNet hybrid offers a practical middle ground: interpretable by construction, competitive at short horizons, and stable at long horizons. The decision framework proposed in Section 6 provides actionable guidance for practitioners.

The full codebase—15 models implemented from scratch, evaluation harness, and Streamlit dashboard—is available at <https://github.com/aminemanai2003/neural-mortality-benchmark>.

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